

TrES-2b

R-1646

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_130916 · created 2026-08-02T07:31:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456551.77912 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.106 +/- 0.037
Transit depth (Rp/Rs)^2	1.12 +/- 0.78 [%]
Orbital Inclination (inc)	84.38 +/- 1.91
Airmass coefficient 1 (a1)	1.024 +/- 0.015
Airmass coefficient 2 (a2)	-0.0163 +/- 0.0072
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	#3 - [43, 339]
Optimal Aperture	3.85
Optimal Annulus	9.39
Transit Duration (day)	0.07 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.106 ± 0.037 vs 0.1254 (deviation 0.48σ)
Duration fit vs published	0.07 d vs 0.0746 d (deviation 0.13σ)
Mid-time O–C	-0.6 min
Depth SNR	2.6
Residual scatter	1.5 %

Light curve

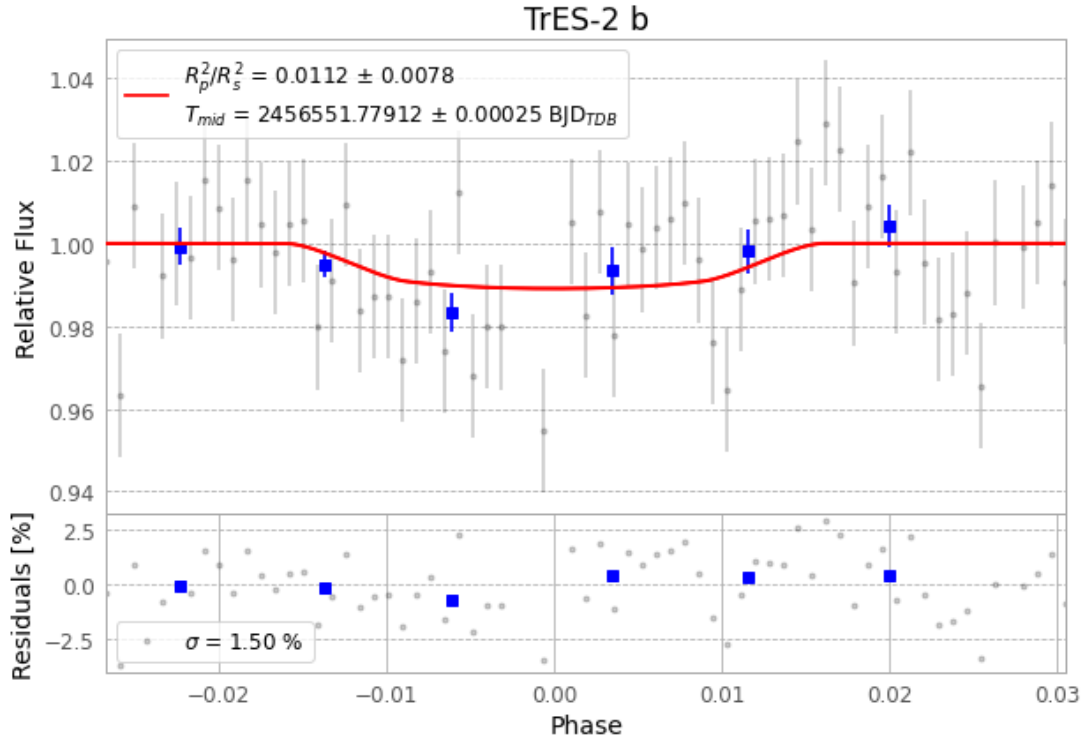


Figure 1 — FinalLightCurve_TrES-2 b_2013-09-15.png (SHA-256 298b5d34a27d86b8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6b42919f80429567...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), TRES-2130916050312.FITS → TRES-2130916082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130916.log (e72d399be45f...), FinalLightCurve_TrES-2 b_2013-09-15.png (298b5d34a27d...), FinalLightCurve_TrES-2 b_2013-09-15.csv (39fa64a8e588...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 07:31Z.